Chenrui Fan

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Google Scholar

Research Overview

My research uses Interpretability as a lens to understand language and multimodal foundation models throughout training and inference, with a focus on Reasoning capabilities and Agent behavior. I study how these capabilities and behaviors emerge, when they fail, and how they can be evaluated, steered, and improved.

Education

- **University of Maryland, College Park** — College Park, MD
  Ph.D. in Computer Science; Advisor: Prof. Soheil Feizi — *Sep 2026 - Present*
- **University of Maryland, College Park** — College Park, MD
  M.S. in Computer Science; Advisor: Prof. Tianyi Zhou; GPA: 4.00/4.00 — *Sep 2024 - Jun 2026*
- **Huazhong University of Science and Technology** — Hubei, China
  BEng. - Computer Science and Technology; GPA: 3.99/4.00; Ranking: Top5%; — *Sep 2020 - Jun 2024*

Selected Research

Reasoning Behavior across Language and Multimodal Models
When do language and multimodal models fail to reason effectively, and what structure underlies their reasoning?

- **Reasoning failures**
  - **MiP-Overthinking.** Reasoning models spend substantial computation on questions with missing premises yet rarely abstain, underperforming non-reasoning models; the overthinking pattern also transfers through distillation.
  - **Thinking Hard Not Smart.** Despite their strength, reasoning models fail to ration shared test-time compute: they follow presentation order and remain insensitive to question value density.
- **Reasoning structure**
  - **ThinkARM.** By adapting a cognitive-science theory of human problem solving, ThinkARM quantitatively distinguishes reasoning patterns across models and reveals how efficiency interventions alter reasoning structure.
  - **V-REX.** Disentangling visual exploration into planning and following reveals capability gaps hidden by outcome-only evaluation.

Learning and Intervention Dynamics
When does behavior generalize during pretraining, and when do inference-time interventions succeed?

- **Inference-time steering** – **ASTEER.** Steering outcomes are predictable from early hidden states, enabling efficient strength selection before full rollout.
- **Pretraining dynamics** – **MoE Grokking.** Examining MoE routing pathways shows that grokking corresponds to internal structural changes despite converged training loss.

Agent Decision-Making
When should agents use tools, and why do their decisions diverge from need?

- **Tool-use decision-making**
  - **MetaTool.** Most tested LLMs struggle to decide whether to use tools and to select the right tools.
  - **TicToc.** Agents remain poorly aligned with human preferences about when stale context requires a new tool call.
  - **Knowing-Doing Gap.** Tool-need signals are internally decodable, but are not always translated into action.

Selected Papers

- **[COLM 2025] Missing Premise Exacerbates Overthinking: Are Reasoning Models Losing Critical Thinking Skill?** [paper] Reasoning Interpretability: **Chenrui Fan***, Ming Li*, Lichao Sun, Tianyi Zhou
- **[ACL 2026 Oral; SAC Highlight Award] Schoenfeld's Anatomy of Mathematical Reasoning by Language Models** [paper] Reasoning Interpretability: Ming Li*, **Chenrui Fan***, Yize Cheng*, Soheil Feizi, Tianyi Zhou
- **[Preprint 2026] When is Your LLM Steerable?** [paper] Interpretability: **Chenrui Fan**, Yize Cheng, Ming Li, Soheil Feizi, Tianyi Zhou
- **[NeurIPS 2026] Model-Adaptive Tool Necessity Reveals the Knowing-Doing Gap in LLM Tool Use** [paper] Agent Interpretability: Yize Cheng*, **Chenrui Fan***, Mahdi JafariRaviz*, Keivan Rezaei, Soheil Feizi
- **[Preprint 2026] Thinking Hard, Not Smart: Reasoning Models Fail to Ration Test-Time Compute Across Questions** [paper] Reasoning Interpretability: **Chenrui Fan***, Yize Cheng*, Ming Li, Yongyuan Liang, Tianyi Zhou, Soheil Feizi
- **[ECCV 2026] V-REX: Benchmarking Exploratory Visual Reasoning via Chain-of-Questions** [paper] Multimodal Reasoning: **Chenrui Fan***, Yijun Liang*, Shweta Bhardwaj*, Kwesi Cobbina, Ming Li, Tianyi Zhou
- **[ICLR 2026] Grokking in LLM Pretraining? Monitor Memorization-to-Generalization without Test** [paper] Pretraining Interpretability: Ziyue Li, **Chenrui Fan**, Tianyi Zhou
- **[ACL 2026 Findings] Your LLM Agents Are Temporally Blind: The Misalignment Between Tool Use Decisions and Human Time Perception** [paper] Agent: Yize Cheng*, Arshia Soltani Moakhar*, **Chenrui Fan***, Parsa Hosseini, Kazem Faghih, Zahra Sodagar, Wenxiao Wang, Soheil Feizi
- **[NeurIPS 2025] ColorBench: Can VLMs See and Understand the Colorful World? A Comprehensive Benchmark for Color Perception, Reasoning, and Robustness** [paper] Multimodal Reasoning: Yijun Liang, Ming Li, **Chenrui Fan**, Ziyue Li, Dang Nguyen, Kwesi Cobbina, Shweta Bhardwaj, Jiuhai Chen, Fuxiao Liu, Tianyi Zhou

- **[CVPR 2024 Oral] MetaCloak: Preventing Unauthorized Subject-Driven Text-to-Image Diffusion-Based Synthesis via Meta-Learning** [paper] Privacy Multimodal: Yixin Liu, **Chenrui Fan**, Yutong Dai, Xun Chen, Pan Zhou, Lichao Sun
- **[ICLR 2024] MetaTool Benchmark: Deciding Whether to Use Tools and Which to Use** [paper] Agent: Yue Huang, Jiawen Shi, Yuan Li, **Chenrui Fan**, Siyuan Wu, Qihui Zhang, Yixin Liu, Pan Zhou, Yao Wan, Neil Zhenqiang Gong, Lichao Sun

*Equal contribution.

Industrial Experience

- **Tencent, WeChat AI** Sep 2023 - Jan 2024
  Research Intern
  - Conducted research on personalized federated learning for mobile input methods under heterogeneous user data.

Honors and Awards

- NeurIPS 2026 AI-Native Academia Workshop Best Paper Honorable Mention Oct, 2026
- NeurIPS 2026 Top Reviewer Oct, 2026
- ACL 2026 SAC Highlight Award Jul, 2026
- Dean's Fellowship, University of Maryland, College Park 2026
- National Scholarship [**7/348≈2.0%**] Sep, 2021

Technical Skills

- **Programming**: Python, C/C++
- **ML Frameworks**: PyTorch, TensorFlow, Hugging Face Transformers, vLLM
- **Tools**: Git, Weights & Biases, LaTeX
- **Languages**: Chinese (native proficiency), English (professional proficiency)

Teaching & Academic Service

- **Teaching Assistant**: CMSC 470: Introduction to Natural Language Processing, University of Maryland, Spring 2026
- **Reviewer**: NeurIPS (Top Reviewer, 2026), ICLR, CVPR, ECCV, COLM, ACL Rolling Review (ARR)